

WEEK7-PRACTICAL

Task1:

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

```
UW PICO 5.09 File: memory.c Modified
#include <stdio.h>
#include <stdlib.h>

int global_var = 10;

static int static_var = 20;

void code_function()
{
    printf("Address of code function : %p\n", (void *)code_function);
}

int main()
{
    int stack_var = 30;

    int *heap_var = (int *)malloc(sizeof(int));
    *heap_var = 40;

    printf("Address of code function : %p\n", (void *)code_function);
    printf("Address of global variable: %p\n", (void *)&global_var);
    printf("Address of static variable: %p\n", (void *)&static_var);
    printf("Address of stack variable : %p\n", (void *)&stack_var);
    printf("Address of heap variable  : %p\n", (void *)heap_var);

    free(heap_var);

    return 0;
}
```

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```
vishalkarnati@VISHALS-MacBook-Air ~ % cd ~/Desktop/os
vishalkarnati@VISHALS-MacBook-Air os % mkdir week7pract
vishalkarnati@VISHALS-MacBook-Air os % cd week7pract
vishalkarnati@VISHALS-MacBook-Air week7pract % nano memory.c
vishalkarnati@VISHALS-MacBook-Air week7pract % gcc memory.c -o memory
vishalkarnati@VISHALS-MacBook-Air week7pract % ./memory
Address of code function : 0x1041f04f8
Address of global variable: 0x1041f8000
Address of static variable: 0x1041f8004
Address of stack variable : 0x16bc0f0f8
Address of heap variable  : 0x104a599c0
vishalkarnati@VISHALS-MacBook-Air week7pract %
```

Task2:

Make use of the `/proc/[pid]/maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
UW PICO 5.09 File: process.c Modified
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 100;

int main()
{
    int stack_var = 200;

    int *heap_var = malloc(sizeof(int));
    *heap_var = 300;

    printf("Process ID: %d\n", getpid());

    printf("Global address : %p\n", (void *)&global_var);
    printf("Stack address  : %p\n", (void *)&stack_var);
    printf("Heap address   : %p\n", (void *)&heap_var);

    printf("Process is running...\n");

    while (1)
    {
        sleep(1);
    }

    free(heap_var);

    return 0;
}
```

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```
vishalkarnati@VISHALs-MacBook-Air week7pract % ./process
Process ID: 2240
Global address : 0x102b60000
Stack address  : 0x10d2a70f3
Heap address   : 0x1034719a0
Process is running...
```

```
Last login: Thu Sep 17 13:40:36 on ttys001
(eval):8: command not found: >
vishalkarnati@VISHALs-MacBook-Air ~ % cd ~/Desktop/os/week7pract
vishalkarnati@VISHALs-MacBook-Air week7pract % ps
PID TTY TIME CMD
2193 ttys000 0:00.00 -zsh
2240 ttys000 0:00.01 ./process
2271 ttys001 0:00.02 -zsh
```